2.10.56

INDHIRESH S - EE25BTECH11027

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Question

Let two non-collinear unit vectors $\hat{a}$ and $\hat{b}$ form an acute angle. A point **P** moves so that at any time t the position vector $\overrightarrow{OP}$ (where **O** is the origin) is given by $\hat{\mathbf{a}}\cos t + \hat{\mathbf{b}}\sin t$. When **P** is farthest from origin **O**, let M be the length of $\overrightarrow{OP}$ and $\hat{\mathbf{u}}$ be the unit vector along $\overrightarrow{OP}$. Then,

1. $\hat{\mathbf{u}} = \frac{\hat{\mathbf{a}}+\hat{\mathbf{b}}}{|\hat{\mathbf{a}}+\hat{\mathbf{b}}|}$ and $M = (1 + \hat{\mathbf{a}} \cdot \hat{\mathbf{b}})^{\frac{1}{2}}$
2. $\hat{\mathbf{u}} = \frac{\hat{\mathbf{a}}-\hat{\mathbf{b}}}{|\hat{\mathbf{a}}-\hat{\mathbf{b}}|}$ and $M = (1 + \hat{\mathbf{a}} \cdot \hat{\mathbf{b}})^{\frac{1}{2}}$
3. $\hat{\mathbf{u}} = \frac{\hat{\mathbf{a}}+\hat{\mathbf{b}}}{|\hat{\mathbf{a}}+\hat{\mathbf{b}}|}$ and $M = (1 + 2\hat{\mathbf{a}} \cdot \hat{\mathbf{b}})^{\frac{1}{2}}$
4. $\hat{\mathbf{u}} = \frac{\hat{\mathbf{a}}-\hat{\mathbf{b}}}{|\hat{\mathbf{a}}-\hat{\mathbf{b}}|}$ and $M = (1 + 2\hat{\mathbf{a}} \cdot \hat{\mathbf{b}})^{\frac{1}{2}}$

Equation

Given equation:

$$\overrightarrow{OP} = \hat{\mathbf{a}} \cos t + \hat{\mathbf{b}} \sin t \tag{1}$$

From given ,if **P** is farthest from origin , then length of $\overrightarrow{OP}$ is given as M.From this we can say that

$$M = max \left\| \overrightarrow{OP} \right\| \tag{2}$$

Theoretical Solution

Now,

$$\left\|\overrightarrow{OP}\right\| = \sqrt{\left(\overrightarrow{OP}\right)^T \left(\overrightarrow{OP}\right)} \tag{3}$$

$$\left\|\overrightarrow{OP}\right\| = \sqrt{(\hat{\mathbf{a}}\cos t + \hat{\mathbf{b}}\sin t)^T(\hat{\mathbf{a}}\cos t + \hat{\mathbf{b}}\sin t)} \tag{4}$$

$$\left\|\overrightarrow{OP}\right\| = \sqrt{\|\hat{a}\|^2 cos^2 t + \left\|\hat{b}\right\|^2 sin^2 t + 2(\hat{a})^T(\hat{b})(cost)(sint)} \tag{5}$$

$$\left\|\overrightarrow{OP}\right\|^2 = \begin{pmatrix} cost & sint \end{pmatrix} \begin{pmatrix} 1 & (\hat{a})^T(\hat{b}) \\ (\hat{a})^T(\hat{b}) & 1 \end{pmatrix} \begin{pmatrix} cost \\ sint \end{pmatrix} \tag{6}$$

Let

$$\mathbf{x} = \begin{pmatrix} cost \\ sint \end{pmatrix} \quad and \quad \mathbf{A} = \begin{pmatrix} 1 & (\hat{a})^T(\hat{b}) \\ (\hat{a})^T(\hat{b}) & 1 \end{pmatrix} \tag{7}$$

Theoretical solution

Then,

$$\left\|\overrightarrow{OP}\right\|^2 = \mathbf{x}^T\mathbf{A}\mathbf{x} \tag{8}$$

Now we should find the maximum value of $x^T Ax$ such that $\|x\| = 1$

By **Rayleigh-Ritz theorem**, the maximum value of the quadratic form if **x** is a unit vector will be the largest eigenvalue (λ_{max}) of the matrix A. So,

$$max\left\|\overrightarrow{OP}\right\| = \sqrt{\lambda_{max}} \tag{9}$$

Now we will calculate the Eigen value for the matrix A :

$$\left|A - \lambda I\right| = 0 \tag{10}$$

$$\left|\begin{pmatrix} 1-\lambda & (\hat{a})^T(\hat{b}) \\ (\hat{a})^T(\hat{b}) & 1-\lambda \end{pmatrix}\right| = 0 \tag{11}$$

Theoretical Solution

$$(1-\lambda)^2 - ((\hat{a})^T(\hat{b}))^2 = 0 \tag{12}$$

$$1-\lambda = (\hat{a})^T(\hat{b}) \;\; or \;\; 1-\lambda = -(\hat{a})^T(\hat{b}) \tag{13}$$

$$\lambda = 1+(\hat{a})^T(\hat{b}) \;\; or \;\; \lambda = 1-(\hat{a})^T(\hat{b}) \tag{14}$$

It's already given that $(\hat{a})^T(\hat{b}) > 0$($\hat{a}$ and $\hat{b}$ form an acute angle . so,

$$\lambda_{max} = 1+(\hat{a})^T(\hat{b}) \tag{15}$$

From Eq.9

$$max\left\|\overrightarrow{OP}\right\| = \sqrt{1+(\hat{a})^T(\hat{b})} \tag{16}$$

The above equation can be written as

$$max\left\|\overrightarrow{OP}\right\| = \sqrt{1+\hat{a}.\hat{b}} \tag{17}$$

Theoretical Solution

From Eq.2:

$$M = \sqrt{1 + \hat{a}.\hat{b}} \tag{18}$$

Now let us find the value of t for which $\left\|\overrightarrow{OP}\right\|$ is max
With eigenvalue equation,We'll use matrix A and largest eigenvalue λ_{max} such that,

$$\left(A - \lambda I\right) x = 0 \tag{19}$$

$$\begin{pmatrix} -(\hat{a})^T(\hat{b}) & (\hat{a})^T(\hat{b}) \\ (\hat{a})^T(\hat{b}) & -(\hat{a})^T(\hat{b}) \end{pmatrix} \begin{pmatrix} cost \\ sint \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \tag{20}$$

By solving this we'll get

$$cost = sint \tag{21}$$

We already know that:

$$sin^2 t + cos^2 t = 1 \tag{22}$$

Theoretical Solution

So,

$$sint = \frac{1}{\sqrt{2}} \quad and \quad cost = \frac{1}{\sqrt{2}} \tag{23}$$

From above result

$$t = \frac{\pi}{4} \tag{24}$$

Now unit vector $\hat{u}$ along $\overrightarrow{OP}$ is given by:

$$\hat{\mathbf{u}} = \frac{\overrightarrow{OP}}{\left\|\overrightarrow{OP}\right\|} \tag{25}$$

$$\hat{\mathbf{u}} = \frac{\hat{\mathbf{a}}\cos t + \hat{\mathbf{b}}\sin t}{\left\|\hat{\mathbf{a}}\cos t + \hat{\mathbf{b}}\sin t\right\|} \tag{26}$$

Theoretical Solution

Now subtituiting the value of t in above equation:

$$\hat{\mathbf{u}} = \frac{\hat{\mathbf{a}}\frac{1}{\sqrt{2}} + \hat{\mathbf{b}}\frac{1}{\sqrt{2}}}{\left\|\hat{\mathbf{a}}\frac{1}{\sqrt{2}} + \hat{\mathbf{b}}\frac{1}{\sqrt{2}}\right\|} \tag{27}$$

$$\hat{\mathbf{u}} = \frac{\hat{\mathbf{a}} + \hat{\mathbf{b}}}{\left\|\hat{\mathbf{a}} + \hat{\mathbf{b}}\right\|} \tag{28}$$

From Eq.18 and Eq.28 (a) is correct

C Code

```c
#include <stdio.h>
#include <math.h>

// Dot product of two 2D vectors
double dot(double a[], double b[]) {
    return a[0]*b[0] + a[1]*b[1];
}

// Magnitude of a 2D vector
double magnitude(double a[]) {
    return sqrt(dot(a, a));
}

// Compute max length M and unit vector u using matrix method
void compute(double a[], double b[], double *M, double u[]) {
    double c = dot(a, b); // a  b
    *M = sqrt(1 + c); // largest eigenvalue's sqrt
```

C Code

```
    // Direction = a + b
    double temp[2] = {a[0] + b[0], a[1] + b[1]};
    double norm = magnitude(temp);
    u[0] = temp[0] / norm;
    u[1] = temp[1] / norm;
}
```

Python Code

```
import ctypes
import numpy as np
import matplotlib.pyplot as plt

# Load shared library
lib = ctypes.CDLL(./vec.so) # use vec.dll on Windows

# Define argument & return types
lib.compute.argtypes = [
    np.ctypeslib.ndpointer(dtype=np.double, ndim=1, flags=
        C_CONTIGUOUS),
    np.ctypeslib.ndpointer(dtype=np.double, ndim=1, flags=
        C_CONTIGUOUS),
    ctypes.POINTER(ctypes.c_double),
    np.ctypeslib.ndpointer(dtype=np.double, ndim=1, flags=
        C_CONTIGUOUS)
]
```

Python Code

```
# Example vectors
a = np.array([1.0, 0.0], dtype=np.double)
b = np.array([0.6, 0.8], dtype=np.double)

M = ctypes.c_double()
u = np.zeros(2, dtype=np.double)

# Call C function
lib.compute(a, b, ctypes.byref(M), u)

print(From C library:)
print(M =, M.value)
print(u =, u)

# Plot in same style as attachment
O = np.array([0.0, 0.0])
P = u * M.value
```

Python Code

```
plt.plot([O[0], P[0]], [O[1], P[1]], 'b-', label=Vector OP)
plt.scatter(*O, color=red, s=100, label=O(0,0))
plt.scatter(*P, color=green, s=100, label=fP({P[0]:.2f},{P[1]:.2f
    }))
plt.scatter(u, color=purple, marker=, s=200, label=fu({u[0]:.2f
    },{u[1]:.2f}))
plt.axhline(0, color='black')
plt.axvline(0, color='black')
plt.legend()
plt.title(Figure)
plt.savefig(/media/indhiresh-s/New Volume/Matrix/ee1030-2025/
    ee25btech11027/MATGEO/2.10.56/figs/figure1.png)
plt.show()
```

Plot

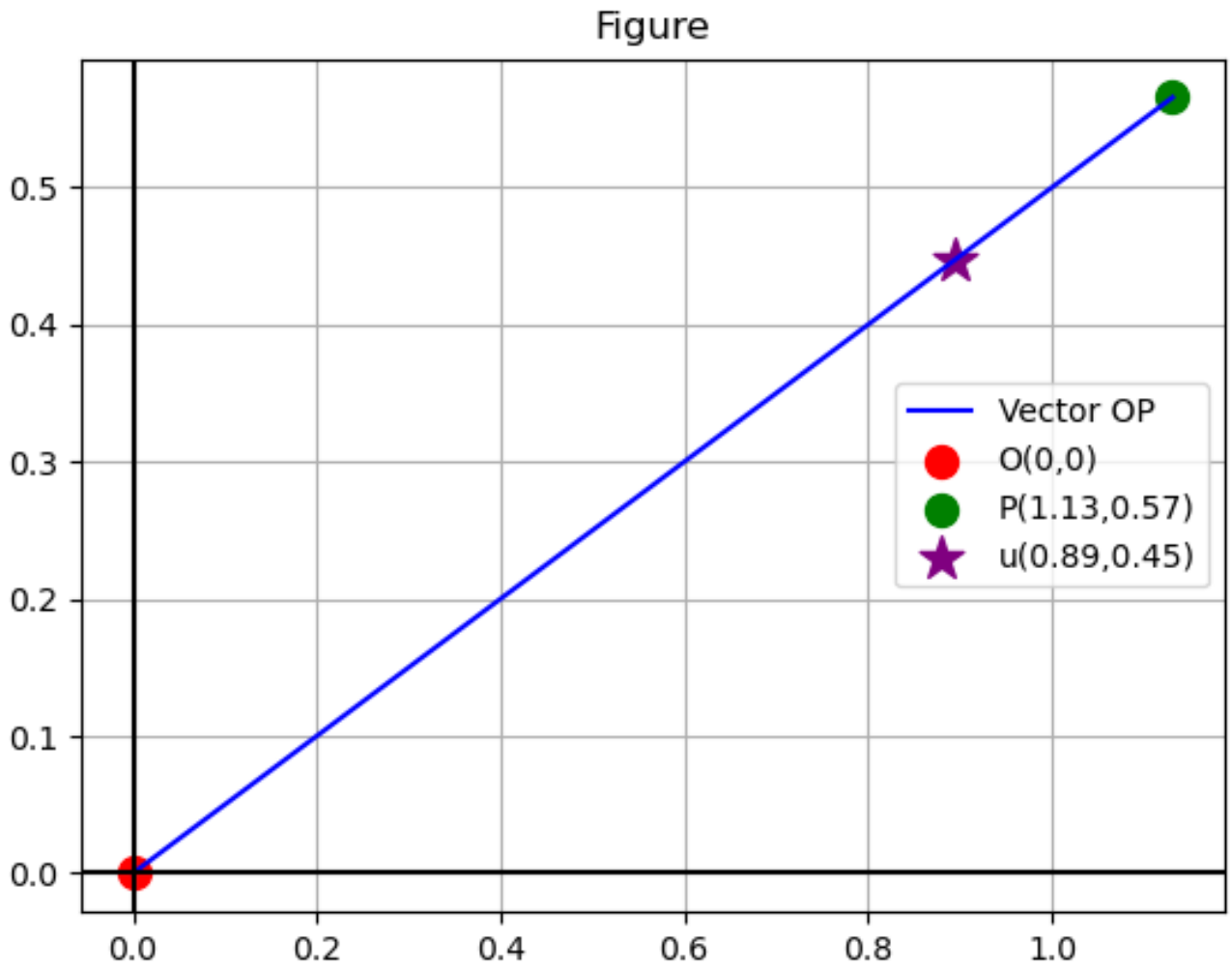
Figure
0.5
0.4
0.3
0.2
0.1
0.0
0.0
0.2
0.4
0.6
0.8
1.0
Vector OP
O(0,0)
P(1.13,0.57)
u(0.89,0.45)